

Answer Key

Electromagnetic Induction: Induced emf & Current

Given Data:

$$N = 25 \text{ turns}$$

$$A = 1.8 \text{ m}^2$$

$$\Delta t = 0.85 \text{ s}$$

$$B_{\text{initial}} = 0.00 \text{ T} ; B_{\text{final}} = 0.55 \text{ T}$$

Angle: $\vec{B} \perp$ plane (Assume $\theta = 0^\circ$)

1. Magnitude of the Induced emf

According to Faraday's Law, the absolute value of the average induced electromotive force (emf) is given by:

$$|e| = \left| \frac{\Delta \Phi_{\text{total}}}{\Delta t} \right|$$

First, we determine the variation of the magnetic flux $\Delta \Phi_{\text{total}}$. Since only the magnetic field B is varying uniformly:

$$\Phi = N \cdot B \cdot A \cdot \cos(\theta)$$

$$\Delta \Phi_{\text{total}} = N \cdot A \cdot \cos(0^\circ) \cdot (B_{\text{final}} - B_{\text{initial}})$$

Substituting the values:

$$\Delta \Phi_{\text{total}} = 25 \cdot 1.8 \cdot 1 \cdot (0.55 - 0.00)$$

$$\Delta \Phi_{\text{total}} = 45 \cdot 0.55 = 24.75 \text{ Wb}$$

Now, calculate the induced emf:

$$|e| = \frac{24.75}{0.85}$$

$$|e| \approx 29.117 \text{ V}$$

Rounding to significant figures:

$$|e| \approx 29.1 \text{ V}$$

2. Magnitude of the Induced Current

The coil acts as a generator forming a closed circuit. According to Ohm's Law:

$$I = \frac{|e|}{R_{total}}$$

Given that the resistance of the coil is $R = 2.5 \, \Omega$:

$$I = \frac{29.117}{2.5}$$

$$I \approx 11.6468 \, \text{A}$$

The magnitude of the induced current is:

$$I \approx 11.6 \, \text{A}$$